Multi-Source Candidate Data Transformer

Stage 1: Technical Design · one page · no code

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Ingest › Extract › Normalize › Canonicalize › Resolve › Reconcile › Calibrate › Project › Validate

Sources are content-addressed; adapters write typed, source-located Claims to an append-only Evidence Ledger. The profile is a deterministic reduce over the ledger; Project + Validate apply the runtime config; the engine itself never changes.

1 PROBLEM & NORTH STAR

Collapse many messy, overlapping, partly-broken sources into one canonical profile per candidate: a fixed schema, normalized, deduplicated, with per-field provenance, confidence and recorded dissent. One invariant rules every stage, **prefer-empty-over-wrong**: on an unresolved conflict I demote confidence or emit `null` and keep the losing values visible; never fabricate. The four constraints are wired into the architecture, not bolted on.

2 ARCHITECTURE: EVIDENCE LEDGER + PROJECTION

The engine is a deterministic reduce over evidence:

- **Content-addressed sources** (id = hash(bytes+type), computed once per source) → identical sources dedupe; runs reproducible & cacheable.
- Each adapter appends typed **Claims** (`{path, raw, source_id, type, extractor, locator}`) to an append-only **Evidence Ledger** (a lightweight frozen record; just a per-candidate list, not a DB). `locator` (CSV row / JSON-pointer / PDF span) gives exact traceability.
- Every canonical value is a tracked envelope `Value<T> = {value, confidence, sources[], method, alternatives[]}`, so provenance, confidence and conflicts are uniform on every field (the given schema tracks this for skills only).
- `candidate_id` = UUIDv5 over canonical identity keys → deterministic, never random.
- The full internal `CanonicalProfile` is always built; a separate **Projection** layer renders the view, and that split makes the configurable-output twist a no-code feature.

3 SOURCES & EXTRACTION: ≥1 STRUCTURED + ≥1 UNSTRUCTURED

Structured: recruiter CSV; ATS JSON (foreign keys mapped). **Unstructured:** GitHub API (repos→languages→skills); résumé PDF/DOCX (section-aware parse); recruiter .txt, via a closed-vocab gazetteer for skills + bounded regex for dates/emails/phones. **Precision > recall:** unrecognized text is dropped, never guessed, and free-text is weighted as low-trust corroboration. Each adapter is independent & best-effort.

4 SCHEMA & NORMALIZATION: NORMALIZE ≠ CANONICALIZE

Default projection flattens envelopes to a clean, given-compatible shape (bare values + compact provenance + `field_confidence`); richer views via config. A `conflicts[]` log lists every demoted field. Two distinct ops: **Normalize** (format) and **Canonicalize** (semantics: skills, orgs, country).

Field	Canonical	Policy (edge-aware)
dates	YYYY-MM +precision	"present"→null+is_current; ambiguous D/M↔M/D → locale hint else down-precision to YYYY (never guess); 2-digit yr = fixed pivot
phones	E.164	region: in-number > candidate country > source > config; invalid→null (never fabricate CC); ext kept aside; dedupe
country	ISO-3166-1 α-2	alias map (USA→US, UK→GB); region→ISO-3166-2 if sure else free text
emails	lower+NFC, valid	canonical form (strip +tag/dots) = match key; original stored = truth
skills	versioned vocab	alias + deterministic fuzzy fallback; unknown kept canonical=false+low-conf; conf by evidence (declared<inferred<corroborated)
links	canonical handle	https, tracking stripped, classified linkedin/github/ portfolio/other
years_exp	number | null	derived from merged non-overlapping spans (method=derived) vs stated; conflict→lower conf

All normalizers are **pure & total** → (value | null + reason), never throw. Vocab/matrix are versioned → reproducible & explainable.

5 IDENTITY RESOLUTION: CONSERVATIVE

Tiered match keys: **A** email · E.164 · github/linkedin handle, any one merges; **B** name + two of {company | school | location} (or B + fuzzy-name ≥ τ); **C** name alone never merges. Conflicting Tier-A keys → keep separate (assume distinct people). Union-find over high-confidence edges; **candidate pairs are blocked on selective keys (handles / company / school) and formed only when one side lacks a Tier-A key**. Location confirms a match but never seeds a block, so resolution stays near-linear, no $O(n^2)$ (measured linear to 8k candidates). When in doubt, don't merge: a false merge is as dangerous as wrong-but-confident.

6 RECONCILE + CONFIDENCE: HONESTY GATE

Single-valued fields use an evidence-weighted vote: $\mathrm{score}(v) = \Sigma_s\, w(s,f) \cdot q(s,v,f) \cdot \mathrm{recency}(s)$. w = field-aware, versioned source-reliability matrix (visible and tunable, no code change). Corroboration emerges as independent sources sum, with source-origin dedup (no double-counting) + diminishing returns.

Winner = argmax, but conflict-aware: normalized-equivalent values collapse first, so the deterministic tie-break (reliability → lexicographic → source_id) only picks a stable representative among equivalents, never the arbiter between materially-different values.

Honesty gate (core safeguard): when materially-different values remain and the margin < τ (a true tie is margin 0), emit `null` for high-stakes fields (identity/contact) or demote to a floor confidence otherwise (prefer_empty_over_wrong); losers kept in `alternatives[]`. Arbitrary choice between distinct values is never allowed.

Multi-valued: union + entity-level dedup; experience/education deduped by (canonical org + overlapping dates + title similarity), merged recursively. Confidence ∈ [0,1] = clamp(reliability·agreement·normalization − conflict_penalty), fixed-rounded (deterministic), documented as ordinal; overall = importance-weighted mean − conflict/coverage penalties.

7 CONFIGURABLE OUTPUT: THE TWIST

Projection = pure declarative transform: `fields[]={path, from, type, required, normalize}` (`from` supports `emails[0]`, `skills[].name`); plus `include_confidence / _provenance / _alternatives` and `on_missing: null|omit|error`. The config itself is validated against canonical paths+types before the run (bad path/type fails fast). Default schema = the built-in default config.

```
{ "fields": [
   {"path":"full_name", "type":"string", "required":true},
   {"path":"primary_email", "from":"emails[0]",
    "type":"string", "required":true},
   {"path":"phone", "from":"phones[0]", "normalize":"E164"},
   {"path":"skills", "from":"skills[].name",
    "type":"string[]", "normalize":"canonical"}
], "include_confidence": true, "on_missing": "null" }
```

8 ROBUST · DETERMINISTIC · EXPLAINABLE · SCALE

Robust: adapter isolation (bad source = 0 claims, continue); total normalizers; honesty gate; two gates (internal invariants + output-schema); unknown⇒null.

Deterministic: content-addressed inputs; pure reduce over the ledger; no randomness/wall-clock in logic; stable sorts; fixed rounding; versioned matrix/vocab; no LLM in core.

Explainable: ledger locators + per-field score breakdown + alternatives[] + conflicts[] → "why this value?" answerable for every field.

Scale: per-candidate reduce is stateless ⇒ embarrassingly parallel & shard-able by blocking key; blocking near-linear (selective keys only, §5); O(1) vocab lookup; the reference CLI runs an **in-memory batch** (linear-time, bounded per candidate), so thousands run linearly (**measured linear to 8k**). Streaming / out-of-core ingest is descoped (§9).

9 EDGE CASES & HONEST SCOPE

Case	Handling
Conflicting value (phone/company)	evidence vote + honesty gate; alternatives[] kept
Different people, same name	tiered keys; name-only never merges
Corrupt / empty / 404 source	adapter isolated, 0 claims, logged; run continues
Garbage free-text note	gazetteer/regex match only → nothing recognized = 0 claims
Ambiguous date 03/04/21	locale hint else down-precision to year
True tie, distinct values	honesty gate → null (high-stakes); never arbitrary pick
Bad config path / missing field	pre-run config validation; on_missing: null|omit|error
Encoding / mojibake / RTL	chardet decode + NFC; never crash

Deliberately descoped (stated, not hidden): ML / probabilistic linkage (rules instead); LLM extraction (deterministic parsers keep determinism + explainability); full org/skill/geo ontologies (seed + pluggable); live LinkedIn scraping (export fixture, ToS); learned weights (fixed matrix); **streaming / out-of-core ingestion** (in-memory batch suffices at target scale; shard per candidate-group to scale out); UI polish (CLI suffices).

10 SURFACE & TESTS

Thin CLI: `transform --inputs <files…> --config <cfg.json> --out profile.json` (default config if omitted) → schema-valid JSON. **Tests:** normalizer units; golden-profile diff on samples incl. an edge case; identity-merge & honesty-gate (true-tie→null) cases; projection/validation for default + custom config.